

Ibrahim Tareq

JUNIOR SOFTWARE ENGINEER

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Summary

Computer Engineering graduate (Alexandria University, 2026) passionate about building systems that scale. Hands-on experience spans distributed systems and stream-processing pipelines (Kafka, Spark, Kubernetes), full-stack SaaS development (TypeScript, React, Node.js, PostgreSQL), HPC parallel programming (OpenMP, MPI), and applied ML/NLP (PyTorch, LLM fine-tuning with Q-LoRA and DPO). Proven ability to take projects from architecture to deployment — seeking an entry-level software engineering role where I can contribute immediately and grow fast.

Education

Alexandria University, Faculty of Engineering
B.S. in Computer and System Engineering

Alexandria, Egypt
Sept 2021 – Jun 2026

- Grade: **Good** | Military service: postponed until age 28

Projects

Contabo Cloud Portal

 GitHub

Graduation Project — Multi-Tenant Cloud Reseller & CD Platform

2025 – 2026

- Production-grade multi-tenant SaaS: resellers manage Contabo VPS for clients via a built-in Git-based CD pipeline (Heroku-style); three-tier RBAC (Admin → Resellers → Clients) with isolated billing, dashboards, and resource quotas.
- **Tech:** TypeScript, React, Node.js, PostgreSQL, Docker, REST APIs

IoT Weather Station Monitoring Platform

Distributed Systems & Stream Processing

2025

- 10-station distributed pipeline: Kafka ingestion → Kafka Streams anomaly detection → dual persistence (custom BitCask LSM store + Parquet archival) → Kubernetes orchestration; 6 Enterprise Integration Patterns applied.
- **Tech:** Java, Kafka, Kafka Streams, Parquet, Elasticsearch, Kibana, Docker, Kubernetes

TPC-H Data Warehouse & Analytics Pipeline

Data Engineering — OLTP-to-OLAP Pipeline

2025

- End-to-end ELT: TPC-H benchmark (22 tables, 1 GB+) from MySQL into a star-schema warehouse processed with Apache Spark — achieved $\sim 19.5\times$ query speedup over MySQL baseline.
- **Tech:** MySQL, Apache NiFi, Apache Spark, Python, Apache Parquet, SQL

Reverse Time Migration — 3D Seismic Wave Equation Solver

High-Performance Computing & Parallel Programming

2025

- RTM algorithm for geophysical subsurface imaging: C finite-difference stencil ($O(n^3 \cdot t)$) parallelized with OpenMP (7.2x on 8 cores) and MPI (14.1x on 16 cores, multi-node); cache-blocking with halo-exchange overlap.
- **Tech:** C, OpenMP, MPI, HDF5, finite-difference methods, performance profiling

Java-to-Bytecode Compiler

Programming Languages & Compilers — Course Project

2025

- Multi-phase Java compiler in C++: NFA→DFA lexer via Thompson's construction; LL(1) parser from FIRST/FOLLOW sets and predictive parse table with leftmost derivations.
- **Tech:** C++, finite automata, context-free grammars, LL(1) parsing, compiler design, JVM bytecode

NLP Course Labs (4 Labs)

NLP & Machine Learning — Course Labs

2024

- N-gram LMs + Naïve Bayes/LR emotion classifier (94% macro-F1); Word2Vec on CONLL2003 + FFNN vs. HMM/Viterbi NER (9 classes); Transformer & Bi-LSTM seq2seq NMT (French→English, beam search, BLEU).
- Q-LoRA fine-tuning of Qwen 1.5B for code generation (SFT); DPO alignment with β -sensitivity analysis. **Tech:** Python, NumPy, PyTorch, HuggingFace, PEFT, wandb

Distributed Systems Course Work

Distributed Systems — CS432 Labs & Experiments

2025

- **RMI Graph:** Java RMI distributed system; BFS/Dijkstra with concurrent Q/A/D operations. **K-Means:** MapReduce on Hadoop & Spark. **Cassandra:** CAP/quorum across 18+ failure scenarios (6 modes $\times$ 3 RF).
- **Kafka vs. JMS:** Throughput, latency & response-time benchmarks (10K-message medians). **Tech:** Java, RMI, Hadoop, Spark, Cassandra, Kafka, ActiveMQ, JMeter, Docker

Skills

Languages	Java, TypeScript, JavaScript, Python, C/C++, SQL
Web	React, Angular, Node.js, HTML5, CSS3, REST APIs
Distributed	Kafka, Kafka Streams, Apache Spark, Cassandra, Elasticsearch, Kibana
DevOps	Docker, Kubernetes, Linux (CLI & shell scripting), Git, Apache NiFi
Databases	MySQL, PostgreSQL, Oracle DB, Apache Parquet, NoSQL (Cassandra)
Engineering	OOP, data structures & algorithms, system design, ERP systems, debugging
ML / AI	PyTorch, HuggingFace, PEFT (LoRA/Q-LoRA), scikit-learn, NumPy, NLP, LLM fine-tuning
Security	Kali Linux, cryptography, network security, Nmap, Burp Suite, Metasploit
Soft Skills	Critical thinking, problem-solving, teamwork, time management, continuous learning

Experience & Training

Ezz Al Dekheila Steel Co. (EZDK) Alexandria, Egypt
Information Technology Engineer Jul 2025

- Worked hands-on with Oracle E-Business Suite (EBS) and Oracle DB; wrote SQL scripts for data extraction, reporting, and routine data operations across production databases.
- Assisted in diagnosing and resolving infrastructure issues across the plant network; gained practical exposure to enterprise networking fundamentals, server configuration, and hardware troubleshooting in an industrial environment.

Arabian Academy Security Alexandria, Egypt
Cyber Security Engineer Jun 2024 – Sep 2024

- Studied and applied GRC frameworks including ISO 27001, GDPR, and PCI-DSS; conducted risk assessments and contributed to security policy documentation.
- Hands-on practice with SIEM tools for log analysis and threat detection; performed penetration testing exercises using Kali Linux, Nmap, Burp Suite, and Metasploit; administered Active Directory environments.
- Explored threat intelligence workflows and incident response procedures in a structured training environment.

National Institute for Telecommunication Education
Ethical Hacking — Digital Egypt Youth Program Nov 2025 – Jan 2026

- 72-hour certified course (score: **95%**); covered reconnaissance, network scanning, vulnerability analysis, system & web application security, SQL injection, DoS attacks, and basic cryptography.
- Reinforced practical offensive-security skills and deepened understanding of the full ethical hacking lifecycle — from foot-printing through exploitation and reporting.

Microsoft × Coursera
Microsoft Full Stack Developer Professional Certificate Dec 2024

- Completed a structured curriculum covering full-stack development with React and Node.js, REST API design and integration, relational and NoSQL database management.
- Applied security best practices (authentication, authorization, input validation) and DevOps fundamentals including containerization, CI/CD pipelines, and deployment monitoring.

Information Technology Institute (ITI)
Linux Administration (Admin I & II) Jul 2025

- Covered core Linux system administration: user and group management, file permissions, process control, and storage configuration (LVM, RAID, partitioning).
- Configured network interfaces, DNS, DHCP, and SSH; automated routine tasks with Bash shell scripting; practiced system monitoring and log analysis.

Interests

Linux	Linux enthusiast since 2017; recently switched to macOS for the Unix-first experience.
Writing	I write descriptive blogs about Linux — some are on my GitHub and on Medium.
Coding	Solving problems on HackerRank, LeetCode, and Codewars to sharpen algorithmic thinking and coding efficiency.

Languages

English	Professional proficiency(B+)
Arabic	Native proficiency